



ya diggg: Final Hardware Selection & BOM Breakdown

This document finalizes the Bill of Materials (BOM) based on the technicalities and scope defined for the ya diggg.

1. Core Component Selection

Subsystem	Component	Manufacturer	Key Specification
SoC (The Brain)	i.MX 8M Nano	NXP	Quad-core ARM Cortex-A53 @ 1.5G Hz. Low power, high OCR throughput Instagram
Image Sensor	AR1335	onsemi	13MP, 1.1um pixel, 4K video capable, specialized AF for small print E-Cosystems .
Display	LS027B7DH01	Sharp	2.7" Memory LCD, 400x240 1-bit, Ultra-low power (50uW static) Youtube .
Memory	MTFC32GAPALNA	Micron	32GB eMMC 5.1 . High reliability high capacity for offline DB storage
Haptics	G1040001D	Jinlong	10mm Linear Resonant Actuator (LRA) for crisp tactile feedback Ineedmotors .
Power Mgmt	AXP209	X-Powers	High-integration PMU for Li-Po charging and power sequencing

2. Estimated BOM Cost (Low MOQ: 500Units)

Category	Estimated Cost (USD)	Components Included
Core ICs	\$42.00	NXP SoC, PMU, eMMC, RAM, USB Controller
Imaging & AF	\$18.50	AR1335 Sensor, AF VCM, Lens Stack
Display Unit	\$14.00	Sharp Memory LCD + FPC Connector
Mechanicals	\$22.00	Translucent PC Shell, CNC Aluminum Frame, Haptic LRA, Buttons
PCB & Passive	\$12.00	6-layer HDI PCB, Connectors & SMT passives
Battery	\$6.50	1500mAh Li-Po + Protection PCBA
Packaging	\$4.00	Premium cardboard + Recycled paper guide
TOTAL BOM	~\$119.00	<i>Excluding NRE, Assembly and Logistics</i>

3. Technical Integration Notes

- **Lens Choice:** Optimized for a FOV of 65° to allow for full 12 " album cover capture at ~30cm and macro OCR at ~5cm.
- **PCB Constraint:** Target PCB dimensions are 85mm x 50mm, utilizing both sides for component density.
- **Heat Path:** Thermal paste (3.0 W/m·K) connects the SoC directly to the internal aluminum frame [Ptmake](#).
- **USB-C:** Supports USB 2.0 High Speed (480Mbps) for database sync. No Power Delivery (PD) required, standard 5V/1.5A charging

4. Next Technical Steps

- ☐ **Prototyping:** Order the **i.MX 8M Nano EVK** and **Sharp LCD Breakout** to begin firmware integration [Adafruit](#)
- ☐ **Optics Validation:** Verify that the AR1335's autofocus range can resolve 6pt Helvetica font (standard for vinyl credits).
- ☐ **Power Profiling:** Measure actual current draw during continuous OCR "burst" modes to finalize battery safety margins